

ORANGE, Australia

WMO: 957260

Lat: 33.3814S

Lon: 149.1269E

Elev: 948

StdP: 90.44

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.4	-1.2	-6.9	2.4	12.9	-5.0	2.8	9.1	12.2	6.5	10.9	6.0	1.5	190	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.2	31.8	17.1	29.7	16.6	28.0	16.2	19.8	26.0	18.9	25.0	18.2	23.8	5.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	14.6	21.2	17.2	13.8	20.7	16.3	13.0	20.2	60.9	25.7	57.7	25.0	55.2	23.9	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.4	8.4	-4.9	34.3	1.0	2.6	-5.6	36.1	-6.2	37.6	-6.8	39.1	-7.5	40.9		
(5)			-5.3	21.2	1.0	0.9	-6.0	21.8	-6.6	22.3	-7.2	22.8	-7.9	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	20.1	18.9	16.2	12.3	8.1	6.0	5.1	5.9	8.9	11.7	15.0	17.5
	DBStd	5.94	3.51	3.27	2.96	2.92	2.58	2.27	2.16	2.27	3.06	3.45	3.73	3.42
	HDD10.0	563	0	0	1	11	68	121	152	128	56	21	4	1
	HDD18.3	2446	22	28	78	181	316	371	410	384	282	205	113	56
	CDD10.0	1337	314	248	194	81	11	1	1	2	24	75	153	234
	CDD18.3	178	78	43	13	1	0	0	0	0	1	12	31	
	CDH23.3	2149	894	454	154	14	0	0	0	0	1	21	192	418
	CDH26.7	682	339	149	27	1	0	0	0	0	1	51	114	
(14) Wind	WSAvg	4.1	4.3	4.1	4.1	3.4	3.7	4.1	4.3	4.4	4.4	4.2	4.2	4.2
(15) Precipitation	PrecAvg	892	88	82	51	55	66	67	88	97	80	81	71	66
	PrecMax	1393	321	289	137	289	194	163	209	252	165	184	157	181
	PrecMin	364	7	7	0	4	4	2	15	4	26	1	7	1
	PrecStd	215	72	67	38	56	41	38	49	52	36	44	38	48
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	33.8	29.4	25.4	18.7	14.5	14.0	16.5	21.6	25.8	30.9	32.6
		MCWB	17.5	18.7	15.7	14.6	12.1	9.6	8.7	10.0	12.3	14.0	15.8	16.2
	2%	DB	32.4	30.6	27.0	22.4	16.9	13.0	12.2	14.4	19.2	23.3	28.0	29.5
		MCWB	17.4	17.7	15.6	13.6	11.1	9.1	8.2	8.6	11.4	13.2	14.2	15.9
	5%	DB	30.4	28.1	24.9	20.5	15.3	11.7	10.9	12.9	17.3	21.3	25.4	27.7
		MCWB	17.2	16.9	15.5	13.0	10.6	8.8	7.7	8.2	10.8	12.7	14.6	15.6
	10%	DB	28.4	26.0	23.0	18.6	13.9	10.7	9.7	11.4	15.6	19.4	23.0	25.6
		MCWB	16.9	16.6	15.1	12.6	10.2	8.5	6.9	7.5	10.0	12.2	14.2	15.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	21.2	19.8	16.6	14.2	11.5	11.6	11.8	14.4	16.4	18.6	19.6
		MCDB	27.9	28.0	23.6	20.8	15.9	12.6	12.6	14.2	18.7	20.7	24.7	25.5
	2%	WB	19.7	19.9	18.2	15.3	12.6	10.5	9.6	10.2	13.0	15.2	17.4	18.5
		MCDB	26.8	25.8	22.0	19.5	15.2	11.6	10.9	12.8	16.8	20.1	23.4	24.1
	5%	WB	19.0	18.9	17.3	14.2	11.4	9.5	8.2	9.0	11.8	14.1	16.4	17.5
		MCDB	26.3	24.0	21.3	18.4	14.3	11.2	9.9	11.9	15.6	18.6	22.0	23.3
	10%	WB	18.2	17.9	16.3	13.4	10.4	8.6	7.2	7.9	10.8	13.1	15.3	16.5
		MCDB	24.9	22.9	20.7	17.4	13.2	10.4	9.1	10.8	14.2	17.8	20.3	22.7
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	12.1	11.6	11.6	10.2	8.0	7.9	9.1	10.6	11.5	12.4	13.1
		MCDBR	16.1	15.1	14.6	14.1	12.7	10.4	10.5	12.3	13.2	14.7	16.1	15.9
	5% WB	MCWBR	5.5	5.5	6.2	6.8	7.8	7.2	7.0	7.7	7.3	7.4	6.7	6.0
		MCDWR	13.0	11.8	10.9	11.2	10.3	8.4	8.1	10.0	10.8	11.9	12.5	12.5
(40) Clear-Sky Solar Irradiance	taub		0.323	0.323	0.302	0.284	0.271	0.263	0.259	0.266	0.292	0.305	0.321	0.315
		taud	2.549	2.577	2.617	2.637	2.622	2.667	2.666	2.630	2.532	2.521	2.516	2.563
	Ebn at Noon		1014	995	983	950	912	898	920	957	976	1002	1011	1027
		Edh at Noon	109	102	91	79	71	64	67	78	97	107	112	108
(44) All-Sky Solar Radiation	RadAvg		7.41	6.40	5.40	4.27	3.16	2.40	2.73	3.63	5.04	6.22	6.82	7.50
	RadStd		0.43	0.53	0.46	0.34	0.21	0.21	0.19	0.27	0.39	0.53	0.63	0.43

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.44	N/A	N/A	+0.85	N/A	N/A	-2	-85	+185	+98

Nomenclature: See separate page